

Liu Yuchen

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Location: Guangzhou/ Zhuhai / Macao

Research Interests: Computational Social Science; Biostatistics and Public Health; Multimodal Model-Driven Research

Education

MSc in Data Science (Precision Medicine), University of Macau

Sep 2025 – Jun 2027

Faculty of Medicine

Expected graduation: Jun 2027

GPA: 3.44 **Supervisor:** Prof. Luo Jiajun

Core Courses: Data Science Programming Methods (A), Data Analysis and Visualization (A-), Artificial Intelligence Methods in Biomedicine (A-), Practical Machine Learning (A-), Medical and Healthcare Data Analysis

Activities: Media Center, UM Postgraduate Association; Volunteer, 15th National Games of China

Sociology, Huaqiao University

Sep 2021 – Jun 2025

School of Philosophy and Social Development

Average Score: 81.15/100

Selected Courses: Big Data and Computational Social Science (92), Urban Sociology (91), Social Survey and Research (95), Market Research and Analysis (88), Probability Theory and Mathematical Statistics (87)

Activities: Student Union Media Center; Volunteer, 2023 China International Fair for Investment & Trade (CIFIT)

Research Experience

Lifestyle Networks, Depressive Symptoms, and GIS-Based Simulation

Jan 2026 – Present

- Analyzed 105,138 survey records using a 36-node Ledoit–Wolf shrinkage partial-correlation network to characterize associations among lifestyle behaviors, work conditions, demographic factors, and depressive symptoms.
- Compared sex- and age-stratified networks across 595 pairwise edges using bootstrap simultaneous uncertainty bands; assessed robustness through a 60% discovery / 40% holdout design and data-quality and threshold sensitivity analyses.
- Identified 13 direct depressive-symptom associations at $|r| \geq 0.04$; all showed consistent directions in the holdout sample, and 12 remained above the display threshold.
- Extended the behavioral-network structure into a GIS-based agent-based modeling framework to explore spatially targeted interventions and effectiveness–equity trade-offs.

LLM-Assisted ABM Review and Literature Intelligence Workflow

Jul 2026 – Present

- Built and analyzed a deduplicated corpus of 29,139 ABM publications from Web of Science, PubMed, and DBLP, enriched with OpenAlex metadata and abstracts.
- Designed an LLM-assisted pipeline to classify research topics, agent types, methodological relevance, LLM adoption, and LLM roles from titles and abstracts.
- Conducted longitudinal topic analysis, statistical trend testing, co-authorship network analysis, and Louvain community detection to study research evolution and collaboration structures.
- Developing a RAG-based workflow combining semantic retrieval, temporal and citation signals, and evidence synthesis to automate literature reviews and generate structured prompts for downstream research agents.

Other Research Experience

Multi-Omics and Biomedical Data Science

Oct 2025 – Dec 2025

- Developed multi-omics and transcriptomic workflows using GEO, TCGA, and GTEx, with differential expression analysis, WGCNA, GO/KEGG enrichment, machine learning feature selection, survival analysis, and two-sample Mendelian randomization.
- In a gastric cancer and gastrointestinal stromal tumor study, integrated GEO microarray datasets, identified 581 shared hub genes, and used Random Forest and LASSO to select 14 candidate biomarkers, including HSD11B1 and CD44.
- In a cervical cancer study, integrated TCGA-CESC ($N = 309$) and GTEx ($N = 19$) RNA-seq data, retained 14,794 high-confidence genes, evaluated multiple machine learning classifiers, and conducted Kaplan–Meier and multivariable Cox analyses across OS, DSS, and PFI.

Internship Experience

Business Analyst Intern, Shanghai LinkCare Information Technology Co., Ltd.

Aug 2024 – Nov 2024

- Analyzed 50,000+ medical product sales records across 21 provinces using Python and BI tools to support performance monitoring, market trend analysis, and supply-chain optimization.
- Conducted procurement and market research involving government institutions and hospitals to support business decisions.

Research Assistant, HKUST Guangzhou Campus

Jul 2026 – Present

- Conduct LLM-assisted bibliometric research and academic network analysis.
- Develop multi-agent social simulation models based on spatiotemporal data, social media text and news text.

Technical Skills

- **Programming:** Python, R, SQL
- **Computational Methods:** Machine Learning, Network Analysis, Agent-Based Modeling, Natural Language Processing, LLM-Assisted Literature Analysis, RAG
- **Biostatistics and Biomedical Analysis:** Regression, Survival Analysis, Transcriptomics, Multi-Omics Analysis
- **Spatial Analysis:** GIS, Spatial Data Analysis, Spatial Visualization
- **Software:** Microsoft Office Suite, SPSS, Stata
- **Language:** IELTS 7.0 (Listening 8.0, Reading 8.5, Writing 6.5, Speaking 6.0)

Awards & Honors

- Second Prize, Fujian Province Challenge Cup Innovation Competition
- Research project selected for *Leaders' Reference*, Fujian Provincial Government